Discrete Mathematics

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Introductory

What is Discrete Mathematics?

Discrete mathematics is the part of mathematics devoted to the study of discrete (as opposed to continuous) objects.

Calculus deals with continuous objects and is not part of discrete mathematics.

Introduction to the logical and mathematical foundations of computer science.

This is not a course in computer programming, but on fundamental concepts of computing.

What is Discrete Mathematics?

We will stress mathematical problem-solving skills and the use of formal concepts as tools for computer science.

Topics include Logic and Proofs; Sets and Functions; Algorithms; Number Theory; Counting; Relations; Graphs; Trees; Boolean Algebra;……

Kinds of Problems Solved Using Discrete Mathematics[1]

How many ways can a password be chosen following specific rules?

How many valid Internet addresses are there?

What is the probability of winning a particular lottery?

Is there a link between two computers in a network?

How can I identify spam email messages?

How can I encrypt a message so that no unintended recipient can read it?

How can we build a circuit that adds two integers?

Kinds of Problems Solved Using Discrete Mathematics 2

What is the shortest path between two cities using a transportation system?

Find the shortest tour that visits each of a group of cities only once and then ends in the starting city.

How can we represent English sentences so that a computer can reason with them?

How can we prove that there are infinitely many prime numbers?

How can a list of integers be sorted so that the integers are in increasing order?

How many steps are required to do such a sorting?

How can it be proved that a sorting algorithm always correctly sorts a list?

Goals of a Course in Discrete Mathematics

Mathematical Reasoning

Combinatorial Analysis

Discrete Structures

Algorithmic Thinking

Applications and Modeling

Goals of a Course in Discrete Mathematics[1]

Mathematical Reasoning:

- Ability to read, understand, and construct mathematical arguments and proofs.

Goals of a Course in Discrete Mathematics 2

Combinatorial Analysis

- Techniques for counting objects of different kinds.

Goals of a Course in Discrete Mathematics 3

Discrete Structures

- Abstract mathematical structures that represent objects and the relationships between them.
- Examples are sets, relations, graphs, trees , etc.

Goals of a Course in Discrete Mathematics 4

Algorithmic Thinking

- One way to solve many problems is to specify an algorithm.
- An algorithm is a sequence of steps that can be followed to solve any instance of a particular problem.
- Algorithmic thinking includes specifying algorithms, analyzing the memory and time required by an execution of the algorithm, and verifying that the algorithm will produce the correct answer.

Goals of a Course in Discrete Mathematics 5

Applications and Modeling

- It is important to understand the applications of the topics in discrete mathematics and develop the ability to develop new models in various domains.
- Concepts from discrete mathematics have not only been used to address problems in computing, but have been applied to solve problems in many areas such as chemistry, biology, linguistics, geography, business, etc.

Discrete Mathematics is a Gateway Course

Topics in discrete mathematics will be important in many courses that you will take in the future:

- **Computer Science**
- **Mathematics**
- **Other Disciplines**

Discrete Mathematics is a Gateway Course[1]

Topics in discrete mathematics will be important in many courses that you will take in the future:

- **Computer Science**
  - Computer Architecture
  - Data Structures
  - Algorithms
  - Programming Languages
  - Compilers
  - Computer Security
  - Databases
  - Artificial Intelligence
  - Networking
  - Graphics
  - Game Design
  - Theory of Computation
  - ……

Discrete Mathematics is a Gateway Course 2

Topics in discrete mathematics will be important in many courses that you will take in the future:

- **Mathematics**
  - Logic
  - Set Theory
  - Probability Theory
  - Number Theory
  - Abstract Algebra
  - Combinatorics
  - Graph Theory
  - Game Theory
  - Network Optimization
  - …

The concepts learned will also be helpful in continuous areas of mathematics.

Discrete Mathematics is a Gateway Course 3

Topics in discrete mathematics will be important in many courses that you will take in the future:

- **Other Disciplines**
  - You may find concepts learned here useful in courses in economics, linguistics, philosophy, and other departments.

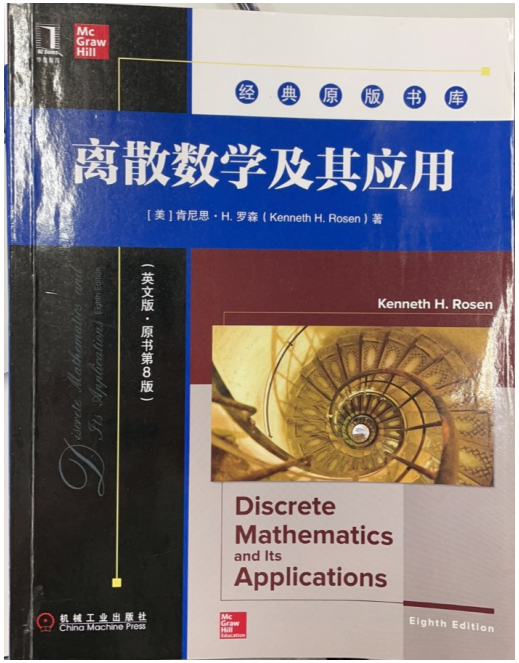
Textbook

Discrete Mathematics and Its Applications Eighth Edition

Author: Kenneth H.Rosen

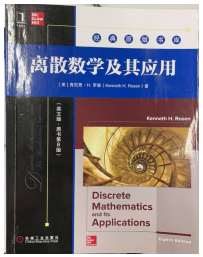
Publisher: China Machine Press; (2020)

ISBN-13: 978-7-111-64530-6



Contents

1. The Foundations: Logic and Proofs
2. Basic Structures: Sets, Functions, Sequences, Sums, and Matrices
3. Algorithms
4. Number Theory and Cryptography
5. Induction and Recursion
6. Counting
7. Discrete Probability
8. Advanced Counting Techniques
9. Relations
10. Graphs
11. Trees
12. Boolean Algebra
13. Modeling Computation



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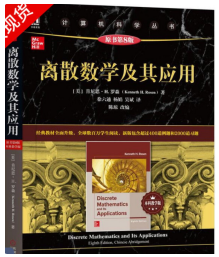
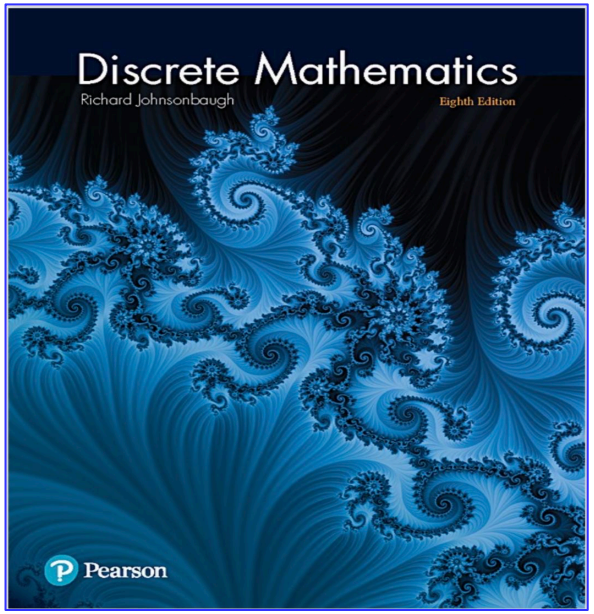
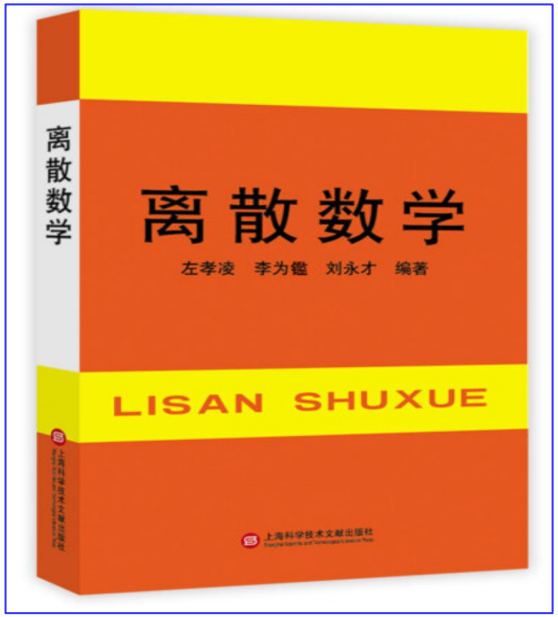


Reference books





Resources

中国大学MOOC

离散数学 国家精品

电子科技大学　王丽杰、傅彦、罗亮、高辉

【课程概述】我们通过对课程内容的合理安排（“营养均衡”），每一讲的精心调配（“正餐”），课后习题的专业配套（“甜点”），为在线学习用户提供了学习离散数学课程的一种新形式。

1903人参加　进行至第1周

离散数学 国家精品

北京交通大学　刘铎

让我们来共同拿起离散数学的柳叶刀，剖析、鉴赏、玩味。体会到思考和解决是生活的快乐之本、幸福之源！作为计算机科学和软件科学的基础，本课程讲解详尽、练习丰富，讨论区无零回复——有问必...

1617人参加　进行至第1周

离散数学概论 国家精品

北京大学　陈斌

【课程概述】p.p1 {margin: 0.0px 0.0px 0.0px 0.0px; font: 12.0px Helvetica} 离散数学是计算机科学的基础理论，离散结构的基础知识和逻辑思维的形式化是信息技术类学生的基本功，...

9388人参加　已结束

Resources

bilibili

Arsdigita Discrete Mathematics（英文字幕）

3968 2019-07-14

总而言之随便取个名字

[MIT] 离散数学及其应用 (双语字幕连载中) Discrete

1.2万 2019-09-14

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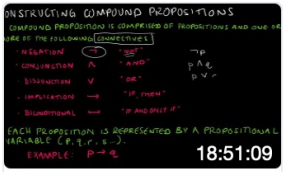

离散数学 Discrete Mathematics 【Rosen】

1279 2021-03-05

泰山椰子

【离散数学及其应用】【离散数学概论】 Discrete

2769 2020-02-20

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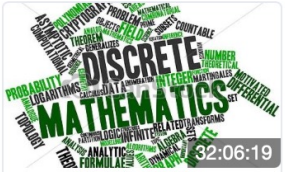
TTC 离散数学 Discrete Mathematics

2104 2019-04-23

负翅与触见



CMU Discrete Mathematics

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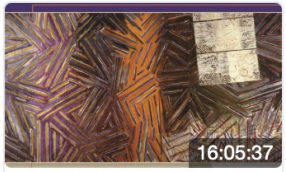

Discrete Mathematics_Greg A.

91 2020-07-25

Sorayya

易志偉老師 - 離散數學 Discrete Mathematics

1312 2020-07-30

Sorayya

Programming Foundations- Discrete Mathematics

135 2020-05-18

法克尤玛德

CS 70: Discrete Mathematics and

2484 2018-04-15

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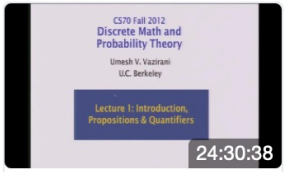

(UC Berkeley离散数学和概率论)CS70 Discrete

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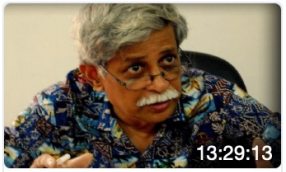

CS 101 - Discrete Mathematics : Spring 2018

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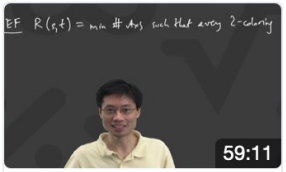

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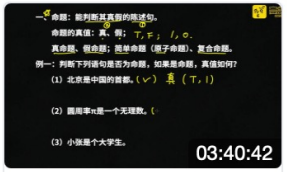

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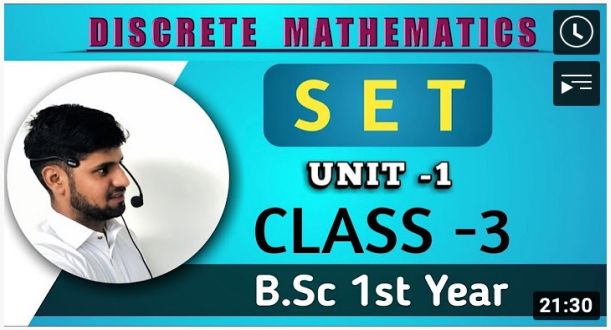

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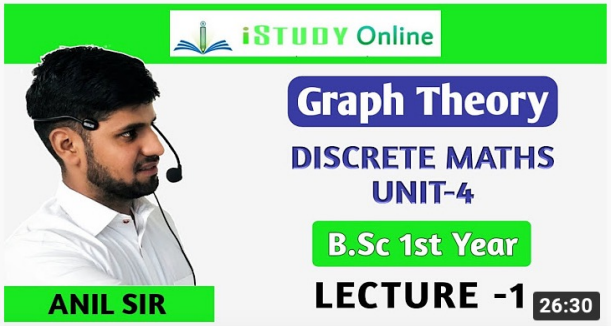

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Final Grades

- Regular grade = 40%
  - Attendance (10%)
  - Homework assignments (20%)
  - Class quizzes (10%)
- Mid-term grade = 30%
- End-term grade = 30%